potentially attractive—but I think an OR Spectrum reviewer could raise several major issues. The most important ones are conceptual rather than cosmetic.

## The criticisms I would expect from reviewers

1. **Major: the paper calls BESO a DRO method, but the implemented optimization model does not appear to be DRO.**
   This is probably the single biggest vulnerability. The paper correctly defines DRO as optimizing the worst expected performance over a family of distributions, and Section 4.2 then constructs

   $$\mathcal{U}_m = \mathrm{conv}\{\hat{P}_n^{b(1)}, \ldots, \hat{P}_n^{b(m)}\}$$

   and writes the corresponding supremum over $P \in \mathcal{U}_m$. orspectrum

   But the model you actually solve in Section 5 is a **scenario-based expected-cost model with equal probabilities**

   $$p_s = \frac{1}{|S|},$$

   and the objective takes a weighted average over scenarios. orspectrum There is no $\sup_{P \in \mathcal{U}}$ in the computational formulation.

   A reviewer may therefore say: *"This is bootstrap-driven stochastic programming/SAA, not distributionally robust optimization."*

   There is even an internal contradiction: Section 4.2 explicitly defines an ambiguity set, whereas the conclusion says BESO works by **"avoiding explicit ambiguity sets."** orspectrum

   **Before submission, I would make a strategic choice:** either genuinely formulate and solve the bootstrap-based DRO problem, or reposition BESO as a **bootstrap-enhanced scenario-based stochastic optimization** framework and considerably soften the DRO claims. The latter may actually make the paper cleaner.

2. **Potentially major/fatal: equations (1)–(3) do not look like the standard Maximum Entropy Bootstrap algorithm you cite.**
   Your Section 4.1 presents "classical MEB" as maximization of discrete Shannon entropy over probabilities $p_i$, subject to a moment constraint, yielding

   $$p_i \propto \exp(\mu f_i),$$

   and then extends this to

   $$p_i^{OA} \propto \exp(\mu f_i + \gamma g_i).$$

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The standard MEB literature describes something materially different: a multi-step time-series resampling algorithm based on **order statistics, intervals, a maximum-entropy piecewise density, random quantiles, and restoration of the temporal ordering**. The official JSS description explicitly presents MEB as a seven-step ensemble algorithm and describes the piecewise-uniform maximum-entropy density and rank-based reconstruction. Journal of Statist... +1

Thus, a statistically sophisticated reviewer may ask whether equations (1)–(3) are actually an MEB formulation at all. I think this issue needs to be resolved explicitly. You could say that OA-MEB is an **entropy-tilting layer applied to MEB-generated candidates**, for example, rather than implying that equation (1) itself is the conventional MEB.

3. **Major: the paper's most novel element, OA-MEB, is introduced theoretically but apparently never implemented or tested.**
   The abstract makes the objective-augmented mechanism a central innovation. orspectrum Yet $\gamma$ and $g_i$ essentially disappear after Section 4.1. The manuscript says $\gamma$ "may be" calibrated using cross-validation, bilevel optimization, or risk preferences, but no calibration rule is actually selected. orspectrum `OA-MEB` itself occurs only in the methodological definition. orspectrum

   A reviewer will very likely ask: **What was $g_i$ in the logistics experiment? What value of $\gamma$ was used? How was it selected? What happens at $\gamma = 0$ versus $\gamma > 0$?**

   I would regard an ablation experiment—ordinary MEB/BESO versus OA-MEB/BESO—as essential if OA-MEB remains a headline contribution.

4. **Major: the theoretical statements in Section 4.2 are much stronger than their justification.**
   In particular, the manuscript states that under "standard bootstrap consistency assumptions," if $m(n) \to \infty$ and $m(n)/n \to 0$, then $\mathcal{U}_{m(n)} \Rightarrow P^\star$, the objective converges uniformly, and the optimizer converges; it then gives a Hausdorff-distance bound. orspectrum

   An OR theory reviewer will ask for the exact assumptions, probability metric/topology, definition of the Hausdorff metric on distributions, regularity of $c(x, \xi)$, meaning of $L$, uniqueness/stability conditions for $x^\star$, and a proof or precise theorem citation. The condition $m(n)/n \to 0$ in particular needs justification rather than being labeled simply "standard bootstrap consistency."

Similarly, $\mathrm{Pen}(m)$ and $\Phi$ appear in equations (4)–(5) without being operationally defined or used later. orspectrum

I would either turn this into a proper theorem/proposition section with assumptions and proofs, or substantially reduce these claims.

5. **Major: the computational study does not support the paper's comparison with DRO.**
   The conclusion says BESO achieves performance "comparable to classical scenario generation and DRO approaches." orspectrum But I cannot find a computational Wasserstein-DRO, moment-DRO, $\phi$-divergence DRO, or even SAA benchmark. Wasserstein and SAA appear in the literature review only. orspectrum +1

   The final operational comparison is BESO against a **Gaussian scenario distribution based on Holt-Winters ETS**, and it is essentially one realized validation instance. orspectrum

   If DRO remains central to the positioning, I would add at least one conventional data-driven DRO benchmark, probably Wasserstein DRO. The standard Esfahani–Kuhn formulation is particularly natural because it constructs a Wasserstein ball around the empirical distribution and provides out-of-sample guarantees. TU Delft Researc...

6. **Major: downstream decision quality needs much stronger out-of-sample validation.**
   The real-life comparison is interesting: ETS gives a cheaper allocation but misses demand for one retailer, while BESO remains feasible. However, the manuscript itself acknowledges that this happened in a **single instance** and "may not have happened." orspectrum

   I would expect a reviewer to request repeated out-of-sample experiments reporting, for example, realized cost, unmet demand, service level, feasibility frequency, regret, and perhaps CVaR/tail cost. One successful Christmas realization is good case-study evidence, but it does not establish general robustness.

7. **The forecasting validation is unusually thin for claims such as "consistently outperforming" competing models.**
   You have 52 series with only 45 monthly observations and a three-month forecast horizon. orspectrum Apparently there is one three-month test period. That means the performance claims rest heavily on one forecast origin. For a forecasting contribution, a reviewer may ask for rolling-origin/time-series cross-validation. Rolling-origin evaluation is standard precisely because it tests multi-step forecasts repeatedly while preserving temporal ordering. OTexts: Online, o... +1

The Friedman/Nemenyi analysis helps, but repeated temporal forecast origins would make the evidence substantially stronger.

8. **There are some concrete methodological inconsistencies a reviewer is likely to spot immediately.**
   One particularly visible one is: *"the parameter $p$ was set ... keeping the one that maximized the corresponding AIC value."* orspectrum Unless you are using an unconventional sign convention, that should almost certainly be **minimized**, not maximized. I would verify the code rather than simply changing the word.

   Another is Table 1: MAE and MSE average ranks are exactly identical for every method in all three bootstrap sizes. orspectrum This is possible, but striking enough that reviewers may wonder whether the same ranking vector was inadvertently reused. I would verify this carefully.

9. **There is an apparent mathematical error in the discussion of boostset size.**
   The manuscript says that *"the number of variables and constraints in the DE formulation do not depend on the size of the boostset."* orspectrum But equations (17)–(23) explicitly introduce $q_{ij}^s$, $\epsilon_{js}$, and constraints for every $s \in S$. orspectrum

   Therefore, if boostset cardinality determines the number of scenarios—as the surrounding text seems to imply—the deterministic equivalent grows with $|S|$. This sentence needs correction or a very clear explanation of the distinction between "boostset size" and "number of optimization scenarios."

10. **The integrality assumptions may conflict with the forecast values.**
    Both $q_{ij}^s$ and $\epsilon_{js}$ are declared integer, while

$$\sum_i q_{ij}^s + \epsilon_{js} = \rho_j^s.$$

    orspectrum AR/ARIMA forecasts are normally real-valued. Unless $\rho_j^s$ is explicitly rounded to integer pallets, this equality is generally impossible. You probably do round somewhere in the implementation, but the paper needs to say exactly how and when.

    Relatedly, $b_j$ is described as a **maximum** number of servers to which client $j$ may be assigned, but the model uses $\sum_i x_{ij} = b_j$, forcing exactly that number of assignments. The text acknowledges the issue, but reviewers may still question whether the operational interpretation corresponds to "allowed to split" versus "must split."

11. **The artificial benchmark instances do not validate BESO itself, despite what the abstract/conclusion suggest.**
    The abstract says the findings are "validated against an extensive set of artificial benchmark instances." orspectrum But Appendix B explicitly says the synthetic study is conducted **only in the deterministic setting** and is intended to assess the computational complexity of the split-allocation problem. orspectrum

    That benchmark validates the optimization formulation/solver behavior, not MEB, OA-MEB, the bootstrap ambiguity mechanism, or BESO's robustness. I would change this claim unless you add synthetic stochastic experiments specifically designed to validate BESO.

12. **There are several citation and presentation errors that should be fixed before reviewers see the manuscript.**
    The most obvious is the sentence attributing the Wasserstein-ball DRO model and finite-sample guarantees to **Ernst and Krishnamoorthy (1998)**. orspectrum Your own reference list shows that Ernst and Krishnamoorthy (1998) is actually a paper on the uncapacitated multiple-allocation $p$-hub median problem. orspectrum The intended reference is clearly Mohajerin Esfahani and Kuhn; their paper develops exactly the Wasserstein-ball formulation and finite-sample guarantees. TU Delft Researc...

    There is also the stray } in the abstract after "performance." orspectrum Beale is listed as 2018 even though the article is from the 1955 volume—the online republication date is 2018, which probably caused the error. OUP Academic And I would systematically check the bibliography because these mistakes can make reviewers less trusting of the more substantive theoretical claims.

## What I think the editor/reviewers are most likely to focus on

The paper's **application is not the weak part**. The real logistics setting, actual 52-store dataset, two-echelon allocation problem, and realized end-of-year comparison are attractive. The problem is that the manuscript currently promises a substantially stronger methodological contribution—**DRO + a new objective-aware MEB + decision-dependent ambiguity + convergence theory**—than the computational study actually demonstrates. The abstract makes those claims particularly prominent. orspectrum

If I were acting as a referee, my tentative recommendation on this version would probably be **major revision**, with a real possibility of rejection if I interpreted points 1–4 as fundamental rather than repairable.

The good news is that I see a fairly clear route to a much stronger submission. I would prioritize, in this order: **(1)** settle whether BESO is genuinely DRO or scenario-based

stochastic programming; **(2)** resolve the relationship between your equations (1)–(3) and genuine MEB; **(3)** actually implement and evaluate OA-MEB/$\gamma$; **(4)** add strong out-of-sample decision experiments and at least one real DRO benchmark; **(5)** either prove or soften the Section 4.2 convergence statements; and **(6)** then clean up the many smaller formulation/citation issues.

If you want, I can next do a **"Reviewer 2" pass** and write the kind of detailed referee report you might actually receive from OR Spectrum—Major Comments, Minor Comments, and a probable recommendation. That would let us identify exactly what I would change before submission.

Sources



**Today** 10:42 AM

thank you, very much to the point. Couldn't I argue that the bootstrap scenarios are empirical realizations of an expected distribution?

Yes — you can absolutely argue that, and I think that is the strongest conceptual defense of what you are doing. The key is to be very precise about **what that argument establishes**.

Your bootstrap forecasts can be viewed as **empirical realizations from an estimated predictive distribution**. In fact, that interpretation is already implicit in the manuscript: the bootstrap procedure generates multiple plausible future paths, and those forecasts are then used as scenarios in the optimization model. orspectrum

The distinction is this:

- If the bootstrap scenarios are treated as draws from **one estimated predictive distribution**, then the resulting optimization is naturally a **scenario-based stochastic program / SAA-type model**.
- If each bootstrap resample is treated as defining a **different empirical probability distribution**, and you optimize against the worst one in a family of such distributions, then you have a genuine **DRO interpretation**.

That second interpretation is actually what Section 4.2 tries to formalize. You define bootstrap empirical measures

$$\hat{P}_n^{b(1)}, \ldots, \hat{P}_n^{b(m)}$$

and then the ambiguity set

$$\mathcal{U}_m = \operatorname{conv}\left\{\hat{P}_n^{b(1)}, \ldots, \hat{P}_n^{b(m)}\right\}.$$

You subsequently define

$$\Psi_m(x) = \sup_{P \in \mathcal{U}_m} E_P[c(x, \xi)].$$

That is a perfectly recognizable DRO construction. orspectrum

So I would defend the **conceptual DRO interpretation** like this:

> The bootstrap does not merely produce arbitrary scenarios. Each bootstrap replication represents a statistically plausible empirical realization of the unknown data-generating process. The collection of bootstrap empirical distributions therefore provides a data-driven approximation to a family of plausible probability distributions, rather than a single assumed distribution.

I think that statement is defensible.

The reviewer problem comes one step later: **your implemented deterministic equivalent does not currently optimize the worst case over those bootstrap distributions.** It assigns the scenarios equal probability,

$$p_s = 1/|S|,$$

and minimizes their expected cost. orspectrum That effectively says:

$$\min_x E_{\hat{P}_{\text{boot}}}[c(x, \xi)],$$

where $\hat{P}_{\text{boot}}$ is one empirical predictive distribution obtained by pooling the scenarios.

DRO would instead require something conceptually like

$$\min_x \sup_{P \in \mathcal{U}_m} E_P[c(x, \xi)].$$

That difference is what a knowledgeable reviewer is likely to seize on.

### There is, however, a nice way to strengthen your position

Because your ambiguity set is the **convex hull of finitely many bootstrap empirical distributions**, you have a particularly simple property:

$$\sup_{P \in \mathrm{conv}\{P_1,\ldots,P_m\}} E_P[c(x,\xi)] = \max_{k=1,\ldots,m} E_{P_k}[c(x,\xi)].$$

The reason is that expectation is linear in $P$, so the worst case over a convex hull is attained at an extreme point.

That means your bootstrap-DRO model could potentially be formulated very naturally as

$$\min_x \max_{k=1,\ldots,m} \left\{ E_{\hat{P}_n^{b(k)}}[c(x,\xi)] \right\}.$$

And that would make the DRO interpretation **much harder to challenge**.

It also gives you a very elegant distinction between three models:

$$\begin{aligned} &\text{Deterministic:} && \min_x c(x,\hat{\xi}), \\ &\text{Bootstrap stochastic:} && \min_x \tfrac{1}{m}\textstyle\sum_{k=1}^m E_{\hat{P}_n^{b(k)}}[c(x,\xi)], \\ &\text{Bootstrap DRO:} && \min_x \max_{k=1,\ldots,m} E_{\hat{P}_n^{b(k)}}[c(x,\xi)]. \end{aligned}$$

That comparison could actually become a **very strong experiment** in the paper.

There is also an important subtlety in your favor: your current model is already more conservative than a plain expected-value model because you require a **common first-stage assignment** $x$ to work across all scenarios, while quantities are scenario-dependent and service shortfalls are heavily penalized. orspectrum So the solution has a robust flavor even though the objective itself is expectation-based. I would call that **scenario robustness**, rather than distributional robustness, unless you add the explicit worst-distribution operator.

So I would not abandon the DRO connection. I would instead phrase it more carefully:

> **BESO uses bootstrap resampling to construct a data-driven family of plausible predictive distributions. This family naturally induces a bootstrap-based ambiguity set. The scenario-based formulation considered here corresponds to optimization under the empirical mixture of these distributions, while a worst-case formulation over the induced ambiguity set provides the corresponding DRO extension.**

That wording would, in my view, resolve a large part of the conceptual vulnerability I raised earlier without giving up your main idea.

And there may be an even better option: **with relatively modest changes to formulation (DE), you may be able to make the computational BESO model genuinely DRO rather than merely DRO-inspired.** I think that is worth exploring, because your Section 4.2 has already done much of the conceptual work.

Sources

very insightful. But an I keep the DE formulation compatible with solvers suchas gurobi or cplex?

Yes. In fact, **for the ambiguity set you already propose in Section 4.2, you can keep a deterministic-equivalent MILP that CPLEX or Gurobi can solve directly.** You do not need a nonlinear or specialized DRO solver.

Your current DE minimizes the equally weighted expected cost over bootstrap scenarios, with one common allocation $x_{ij}$ and scenario-dependent recourse variables $q_{ij}^s$ and $\epsilon_j^s$. orspectrum The common $x$ across all scenarios is already exactly the right structure for a two-stage robust/stochastic formulation. orspectrum

The trick is simply to replace the **average over distributions** by an **epigraph representation of the worst distribution**.

Suppose bootstrap replication $k$ induces an empirical distribution $\widehat{P}_k$, and let

$$C_k(x) = \mathbb{E}_{\widehat{P}_k}[c(x,\xi)]$$

denote the expected optimized cost associated with that bootstrap distribution. Your Section 4.2 defines precisely this kind of finite family and takes its convex hull as the ambiguity set. orspectrum

Then

$$\min_x \sup_{P \in \mathcal{U}_m} E_P[c(x,\xi)]$$

with

$$\mathcal{U}_m = \mathrm{conv}\{\widehat{P}_1, \ldots, \widehat{P}_m\}$$

is equivalent to

$$\min_x \max_{k=1,\ldots,m} C_k(x).$$

And **that max is trivial to linearize**. Introduce one continuous variable, say $z$, and write

$$\min z$$

subject to

$$z \geq C_k(x), \qquad k = 1, \ldots, m.$$

That's it. As long as each $C_k(x)$ comes from your existing linear/MILP recourse structure, the resulting model remains a **MILP**.

### Applied directly to your formulation

For example, suppose bootstrap distribution $k$ contains scenarios $s \in S_k$, with probabilities $p_{ks}$. Define

$$C_k = \sum_{s \in S_k} p_{ks} \left[ \sum_{j \in J} \left( \sum_{i \in I} (c_{ij} x_{ij} + d_i q_{ij}^{ks}) + M \epsilon_j^{ks} \right) \right].$$

Instead of your present objective

$$\min \sum_s p_s (\cdots),$$

you could formulate

$$\boxed{\min z}$$

subject to

$$\boxed{z \geq \sum_{s \in S_k} p_{ks} \left[ \sum_{j \in J} \left( \sum_{i \in I} (c_{ij} x_{ij} + d_i q_{ij}^{ks}) + M \epsilon_j^{ks} \right) \right], \quad k = 1, \ldots, m}$$

and retain the scenario constraints in essentially the same form:

$$\sum_i q_{ij}^{ks} + \epsilon_j^{ks} = \rho_j^{ks},$$

$$\sum_j q_{ij}^{ks} \leq Q_i,$$

$$q_{ij}^{ks} \leq \rho_j^{ks} x_{ij},$$

with the same binary $x_{ij}$, integer $q_{ij}^{ks}$, etc.

**Gurobi and CPLEX see nothing exotic here.** They see binary/integer variables, continuous variables, linear equalities, linear inequalities, and one additional continuous variable $z$.

### There is an important modeling choice, though

You need to decide what constitutes **one bootstrap distribution**.

At present your manuscript appears to treat each forecast realization essentially as a scenario and assigns

$$p_s = \frac{1}{|S|}.$$

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If you simply say that **every individual scenario is one distribution**, the DRO model becomes

$$\min_x \max_s c(x, \rho^s),$$

which is basically a worst-scenario robust model. That could be **much more conservative than what you want**.

I think the more attractive construction for BESO is hierarchical:

$$\text{historical data} \longrightarrow \underbrace{\widehat{P}_1, \widehat{P}_2, \ldots, \widehat{P}_m}_{\text{bootstrap predictive distributions}} \longrightarrow \underbrace{\text{scenarios within each } \widehat{P}_k}_{\text{forecast realizations}}.$$

Then you take the **expectation within each bootstrap distribution**, but the **worst case across bootstrap distributions**:

$$\boxed{\min_x \max_{k=1,\ldots,m} \mathbb{E}_{\widehat{P}_k}[c(x, \xi)]}$$

That is much more interesting scientifically than simply minimizing the worst individual forecast.

It also gives a clean interpretation:

> uncertainty **within** a bootstrap distribution is handled stochastically through expected recourse, while uncertainty **about which bootstrap distribution best represents the unknown data-generating process** is handled distributionally robustly.

That is a strong conceptual story.

### And it fits your existing Section 4.2 beautifully

You already write

$$\Psi_m(x) = \sup_{P \in \mathcal{U}_m} E_P[c(x, \xi)]$$

for the convex hull of bootstrap empirical distributions. orspectrum Because expectation is linear in $P$, the supremum over that convex hull occurs at an extreme point. Consequently,

$$\sup_{P \in \mathcal{U}_m} E_P[c(x,\xi)] = \max_{k=1,\ldots,m} E_{\widehat{P}_k}[c(x,\xi)].$$

So your theoretical Section 4.2 would then correspond **directly to the model actually solved** rather than merely providing a DRO interpretation around an expectation-based DE.

### It would also give you a very nice computational experiment

You could compare exactly the same underlying bootstrap information under three prescriptions:

$$\begin{aligned}
\text{Deterministic:} \quad & \min_x c(x, \bar{\rho}), \\
\text{BESO-SP:} \quad & \min_x \frac{1}{m} \sum_{k=1}^{m} E_{\widehat{P}_k}[c(x,\xi)], \\
\text{BESO-DRO:} \quad & \min_x \max_{k=1,\ldots,m} E_{\widehat{P}_k}[c(x,\xi)].
\end{aligned}$$

And then evaluate all three on the **same unseen realized demand**.

That would answer almost automatically the reviewer question: *"What exactly does distributional robustness buy you?"*

You could report something like cost, unmet demand, leased capacity, and out-of-sample feasibility for all three. The computational burden increases because you replicate recourse variables for the distributions/scenarios, but **the mathematical class does not change: it remains a MILP directly solvable by CPLEX/Gurobi**.

There is also a useful intermediate possibility if the pure $\max_k$ turns out to be conservative: you can construct an ambiguity set over the **weights assigned to the bootstrap distributions** and dualize the inner LP. In many cases that **also remains an MILP/LP deterministic equivalent**. That might actually give BESO a very elegant tunable robustness parameter between your present expected-value formulation and full worst-case BESO-DRO.

Sources